United States Patent	12398622
Kind Code	B2
Date of Patent	August 26, 2025
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Systems and methods for delivering fluid into a wellbore

Abstract

A valve assembly for delivering a fluid into a wellbore includes a stationary valve assembly that is installed on a production tube that extends down into the wellbore. A dynamic valve assembly that is attached to a capillary line is lowered into the production tube, and the dynamic valve assembly releasably latched onto the stationary valve assembly. The latching of the dynamic valve assembly to the stationary valve assembly also opens a fluid connection between the dynamic valve assembly and the stationary valve assembly, thereby allowing pressurized fluid to flow from the capillary line into the wellbore via the stationary valve assembly. When it is necessary or desirable to perform maintenance or repair operations on the dynamic valve assembly or the capillary line, the dynamic valve assembly and the attached capillary line can all be withdrawn to the top of the well.

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Appl. No.: **18/779708**

Filed: **July 22, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20250188819 A1	Jun. 12, 2025

Related U.S. Application Data

continuation-in-part parent-doc US 18536887 20231212 US 12044105 child-doc US 18779708

Publication Classification

Int. Cl.: **E21B34/14** (20060101); **E21B33/12** (20060101)

U.S. Cl.:

CPC **E21B34/14** (20130101); **E21B33/1208** (20130101);

Field of Classification Search

CPC: E21B (34/14); E21B (33/1208)

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Background/Summary

(1) This application is a continuation-in-part of application Ser. No. 18/536,887, filed Dec. 12, 2023, the content of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) The invention is related to systems and methods for delivering a fluid chemical into a wellbore. Typically, a capillary line is run from the top of the well down into a desired location or depth within the well. A fluid is then pumped down the capillary line so that the fluid is emitted from the end of the capillary line. In many instances, an injection valve is attached to the end of the capillary line. The injection valve operates to control the rate at which fluid is emitted and to prevent fluids

within the wellbore from entering the capillary line.
(2) The pressure of the fluid within a wellbore increases with depth. The pressure of the fluid in the capillary line must be greater than the pressure of the fluid within the wellbore at the depth at which the fluid is emitted from the capillary line. As a result, it is necessary to use a pump to pressurize the fluid delivered into the capillary line so that the fluid can be emitted from the end of the capillary line.
(3) Typically a production tube extends down into the wellbore. A capillary line used to deliver a fluid into a wellbore is often attached to the outer surface of the production tube via clips, bands or some other sort of attachment mechanism. Unfortunately, it is common for corrosion to occur at the interface between the capillary line and the exterior of the production tube. The corrosion can occur due to the contact between the attachment mechanism, the capillary line and the production tube, due to the pressure applied by a clamp in order to attach the capillary line to the production tube, and as a result of a cathodic corrosion which can occur when two different materials such as carbon steel and chromium steel are brought into engagement. The engagement between the capillary line the attachment mechanism and the production tube also can limit access of a corrosion inhibitor, making corrosion more of an issue. Moreover, the wellbore itself is typically a corrosive environment and is typically filled with various chemicals that exist at high pressures and elevated temperatures. All of these effects can result in a capillary line or tubing leaking or becoming clogged or damaged over time.
(4) When production tubing fails or becomes impaired, it is difficult and expensive to replace. Similarly, when the capillary line is attached to the production tube, it is impossible to conduct any sort of maintenance on an injection valve attached to the end of the capillary line without removing the production tube.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a sectional view of a stationary valve assembly;
(2) FIG. **1**B is a sectional view of a stationary valve assembly mounted to a production tube;
(3) FIG. **1**C is a bottom view of a seating nipple mounted to the end of a production tube;
(4) FIG. **2** is a sectional view of a dynamic valve assembly;
(5) FIG. **3** is an enlarged sectional view of a portion of a dynamic valve assembly;
(6) FIG. **4** is a sectional view illustrating both a dynamic valve assembly and a stationary valve assembly mounted in a production tube;
(7) FIG. **5** is a sectional view of a portion of a dynamic valve assembly taken along section line **5-5** in FIG. **4**;
(8) FIG. **6** is a sectional view of a portion of a dynamic valve assembly taken along section line **6-6** in FIG. **4**;
(9) FIGS. **7**A-**7**H are sectional views that illustrate how a dynamic valve assembly latches to a stationary valve assembly within a production tube;
(10) FIG. **8** is a sectional view of a portion of a production tube with a special seating coupler installed therein;
(11) FIG. **9** is a sectional view of the portion of a production tube depicted in FIG. **8** after a dart assembly has been mounted in the seating coupler;
(12) FIG. **10** is a sectional view of a dynamic valve assembly that is configured to couple to a stationary valve assembly mounted to the bottom portion of a production tube;
(13) FIG. **11** is a sectional view illustrating the dynamic valve assembly depicted in FIG. **10** coupled to a dart assembly that has been mounted in the seating coupler as depicted in FIG. **9**;
(14) FIG. **12** is a sectional view illustrating another embodiment where elements to include a dart

assembly and an injection valve have been mounted in a seating coupler installed in a section of a production tube;
(15) FIG. **13** is a sectional view of a spear that comprises part of an assembly for delivering a fluid into a well;
(16) FIG. **14** is a sectional view of a dynamic assembly that includes the spear depicted in FIG. **13**;
(17) FIG. **15**A is a sectional view of a flow plug assembly that can be part of an assembly for delivering a fluid into a well;
(18) FIG. **15**B is a sectional view of the flow plug assembly illustrated in FIG. **15**A in an assembled condition;
(19) FIG. **16** is a sectional view of a flow plug assembly and a spear, which together form part of an assembly for delivering a fluid into a well;
(20) FIG. **17** is a sectional view of a flow plug assembly that includes a cylindrical shutter which forms part of an assembly for delivering a fluid into a well;
(21) FIG. **18**A is a sectional view of the combination of a spear and a flow plug as depicted in FIG. **17**, where the spear is partially inserted into the flow plug;
(22) FIG. **18**B is a sectional view of the flow plug and spear of FIG. **18**A with the spear fully inserted into the flow plug;
(23) FIGS. **19**A and **19**B are sectional views showing the combination of a spear and a second embodiment of a flow plug, with the spear partially inserted into the flow plug in FIG. **19**A and the spear fully inserted into the flow plug in FIG. **19**B;
(24) FIG. **20** is a sectional view of another embodiment of a spear which can be part of an assembly for delivering a fluid into a well;
(25) FIG. **21** is a sectional view of another embodiment of a flow plug that can be used in conjunction with the spear illustrated in FIG. **20** to form an assembly for delivering a fluid into a well;
(26) FIG. **22**A shows the flow plug of FIG. **21** and the spear of FIG. **20** partially inserted into the flow plug;
(27) FIG. **22**B shows the flow plug and spear of FIG. **22**A with the spear fully inserted into the flow plug;
(28) FIG. **23** is a sectional view of another embodiment of a flow plug which includes a valve;
(29) FIG. **24** is a sectional view of another embodiment of a spear including a cylindrical shutter mechanism;
(30) FIG. **25** is a sectional view of a flow plug which can be used in conjunction with the spear illustrated in FIG. **24** to form an assembly for delivering a fluid into a well;
(31) FIG. **26**A shows the flow plug of FIG. **25** and the spear of FIG. **24** with the spear partially inserted into the flow plug;
(32) FIG. **26**B shows the flow plug and spear combination of FIG. **26**A with the spear fully inserted into the flow plug; and
(33) FIG. **27** is a sectional view of a flow plug and spear combination which includes a bidirectional check valve.
DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS
(34) The following detailed description of preferred embodiments refers to the accompanying drawings, which illustrate specific embodiments of the invention. Other embodiments having different structures and operations do not depart from the scope of the present invention.
(35) The systems and methods disclosed herein allow for a fluid capillary line attached to a dynamic valve assembly to be run down the interior of a production tube and to be releasably latched onto a stationary valve assembly mounted on the production tube. While the dynamic valve assembly is latched to the stationary valve assembly, fluid within the capillary line can be delivered through the dynamic valve assembly into the stationary valve assembly. The stationary valve assembly can then deliver the injection fluid to a location outside the production line.

(36) When it is necessary or desirable to perform maintenance or repair operations on the capillary line or the dynamic valve assembly, the dynamic valve assembly can be unlatched from the stationary valve assembly, and the capillary line and the attached dynamic valve assembly can be withdrawn up to the top of the well and removed from the production tube.
(37) This type of arrangement eliminates the need to clamp a capillary line to the exterior production tubing. As a result, corrosion caused by clamping the capillary line to the production tube or due to cathodic or fretting reaction is largely eliminated. This also makes it possible to perform maintenance and repair operations on the capillary line and the attached dynamic valve assembly without the need to remove the production tube from the well.
(38) FIG. **1**A illustrates elements of a stationary valve assembly **100** that would be attached to a production tube that extends down into a wellbore. The stationary valve assembly **100** could be attached to the bottom of the production tube, or possibly to an interim portion of the production tube that is located part of the way down the full depth of a wellbore. FIG. **1**B illustrates the stationary valve assembly attached to and end of a production tube **117** such that most of the stationary valve assembly is located inside the production tube **117**.
(39) The stationary valve assembly **100** includes a seating nipple **109** that is secured to the end of the production tube **117** via a collar **115**. The seating nipple **109** includes an axial fluid passageway **104** which connects with a radial fluid passageway **106** that extends out to the side of the seating nipple **109**. Fluid delivered to the upper end of the axial fluid passageway **104** can be emitted through the radial passageway **106**, which ensures that the fluid is delivered outside the production tubing **117**.
(40) One or more vent holes **108** can be cut or formed through the length of the seating nipple **109** so that fluid and particles inside the production tube **117** can fall out the bottom of the production tube **117** via the vent holes **108**. This helps to prevent particles from collecting at the bottom of the production tube **117**, which could interfere latching a dynamic valve assembly to the stationary valve assembly. FIG. **1**C illustrates the bottom of the seating nipple **109**, which illustrates how the vent holes **108** pass through the length of the seating nipple **109**. In the embodiment illustrated in FIGS. **1**A-**1**C, the bottom portion of the seating nipple **108** includes a cylindrical wall that forms a vent passageway **103** to shield the vent holes **108**, and to ensure that the vent holes remain open.
(41) The stationary valve assembly also includes an injection valve **110**. The injection valve includes a threaded rear connector **116** at the lower end of the injection valve **110** which is screwed into a threaded receiving connector **102** at the top of the seating nipple **109**. The injection valve **110** essentially operates as a check valve, allowing fluid to flow down into the seating nipple **109** and then out into the wellbore, but preventing fluid located in the axial flow passageway **104** of the seating nipple **109** from flowing upward into a capillary line when the capillary line is latched onto the seating nipple assembly.
(42) A dart **120** is connected to an upper end of the injection valve **110**. In some embodiments, a threaded rear connector **121** on the dart **120** is screwed into a receiving threaded hole **112** at the top of the injection valve **110**. The dart **120** operates to open or close a valve located on a dynamic valve assembly, as will be explained in greater detail below.
(43) The dart **120** includes an axial fluid passageway **125** which extends along a portion but not all of the length of the dart **120**. The axial fluid passageway **125** is connected to radial fluid passageways **126** which extend outward to the exterior surface of the dart **120**. The upper end of the dart includes a tip **127** which is used to open a valve on a dynamic valve assembly, as described below. In addition, the dart **120** includes an outwardly protruding annular ridge **122** which includes both a leading shoulder **123** and a trailing shoulder **124**.
(44) FIG. **2** illustrates a dynamic valve assembly **200** which would be attached to the bottom of a capillary line which is to be run down the interior of a production tube **117**. The dynamic valve assembly is configured to removably latch onto a stationary valve assembly, such as the one illustrated in FIGS. **1**A-**1**C.

(45) The dynamic valve assembly **200** includes a housing **208**, which includes a fluid receiving passageway **231** and a fluid delivery passageway **205** that are separated by a valve seat **214**. A valve element, which in this embodiment is a ball **212**, is movably mounted within the receiving fluid passageway **231**. The lower end of the housing can include a seating shoulder **219** which is configured to bear against the leading edge **123** of the annular ridge **122** of the dart **120** when the dart **120** is inserted into the housing **208**, as will be described in more detailed below. In some embodiments, an elastic or cushioning member may be inserted into the housing **208** and act as the seating shoulder **219**.
(46) FIG. **3** provides an enlarged view of a portion of the housing **208** of the dynamic valve assembly **200**. As shown in FIG. **3**, the valve seat **214** is located between the fluid receiving passageway **231** and the fluid delivery passageway **205**. A biasing element **213**, which in this embodiment is a spiral spring, urges the ball **212** into engagement which the valve seat **214** to keep the fluid receiving passageway **231** isolated from the fluid delivery passageway **205**.
(47) As illustrated in FIG. **2**, a housing connector **209** couples a cap **216** to the upper end of the housing **208**. The cap **216** includes an axial fluid passageway **215** which extends down the central longitudinal axis of the cap **216**. Two rear seals **217** form a fluid tight seal between the exterior cap **216** and the interior of the housing **208**.
(48) A weight bar **226** is attached to the upper end of the cap **216**. The weight bar **226** also includes an axial fluid passageway **225** which extends down its longitudinal axis. The purpose of the weight bar **226** is to provide weight at the end of the capillary line to which the dynamic valve assembly **200** is attached. The weight of the weight bar **226** provides a downward force that is used to help latch the dynamic valve assembly **200** to a stationary valve assembly, as will be described in greater detail below. Centralizers **230** can be located around various portions of the dynamic valve assembly **200**, including the housing **205** and the weight bar **226**. The centralizers **230** help to keep the dynamic valve assembly **200** centered in the production tube **117** through which it is run.
(49) The lower end of the dynamic valve assembly **200** includes a collet **202** which is attached to a lower end of the housing **208**. The collet **202** includes a plurality of prongs **203** which are sufficiently flexible that they can bend outwards and then return to the positions illustrated in FIG. **2**. The prongs **203** of the collet **202** are used to help latch the dynamic valve assembly **200** to a stationary valve assembly, as will be described below.
(50) One or more forward seals **210** are also provided on the interior of the lower section of the housing **208**. The lower seals **210** act to form a fluid tight seal between the interior of the housing **208** and an exterior of a dart on a stationary valve assembly, as is described below.
(51) The upper end of the weight bar **226** of the dynamic valve assembly **200** is attached to a bottom end of a capillary line. The capillary line can then deliver pressurized fluid into the axial flow passageway **225** of the weight bar **226**, through the axial flow passageway **215** of the cap **216** and into the receiving fluid passageway **231** of the housing **208**. The ball **212** pressed against the valve seat **214** of the housing **208** by the spring **213** prevents the pressurized fluid from escaping the housing **208** until the dynamic valve assembly **200** has been latched onto a stationary valve assembly.
(52) Once the dynamic valve assembly **200** has been attached to a capillary line that delivers pressurized fluid, the capillary line and the attached dynamic valve assembly **200** are lowered down a production tube **117** that extends down into a wellbore. The bottom of the dynamic valve assembly **200** is then lowered onto and latched to a stationary valve assembly **100** mounted on the production tube **117**, such as the one illustrated in FIG. **1**B. This involves lowering the dynamic valve assembly **200** such that the housing **208** receives the dart **120** at the top of the stationary valve assembly **100**. The prongs **203** of the collet **202** interact with the annular ridge on the dart **120** to releasably latch the dynamic valve assembly **200** to the stationary valve assembly **100**.
(53) Latching the dynamic valve assembly **200** onto the dart **120** at the top of the stationary valve assembly **100** causes the valve within the housing **208** of the dynamic valve assembly **200** to open

so that fluid in the receiving fluid passageway **231** of the housing can flow into the axial fluid passageway **125** of the dart **120** via the radial fluid passageways **126** of the dart **120**. The latching process, which results in the valve opening, is described in greater detail below.

(54) FIG. **4** provides a cross-sectional view of a dynamic valve assembly **200** latched onto a stationary valve assembly **100** inside a production tube **117**. FIG. **5** is a sectional view taken along section line **5-5** of FIG. **4**. FIG. **5** illustrates the centralizer **230**, which surrounds a portion of the dynamic valve assembly **200**, keeping the dynamic valve assembly **200** essentially centered within a production tube **117**. FIG. **5** also illustrates the cap **216** and the axial flow passageway **215** within the cap **216**.

(55) FIG. **6** is a sectional taken along section line **6-6** of FIG. **4**. FIG. **6** illustrates the tip **127** of the dart **120** located inside a portion of the housing **208**. This view also illustrates that the collet **202** surrounds the exterior of the housing **208**.

(56) FIG. **7**A-**7**H illustrate how the dynamic valve assembly **200** as illustrated in FIG. **2** is releasably latched onto the top of a stationary valve assembly **100** as illustrated in FIGS. **1**A-**1**C. This would occur as the dynamic valve assembly **200** attached to the bottom of a capillary line is gradually lowered through the interior of a production tube **217**. The actual latching operation requires a certain amount of force to push the bottom of the dynamic valve assembly **200** into engagement with the top of the stationary valve assembly **100**. That downward force is applied by the weight of the dynamic valve assembly **200**, as well as the weight of the capillary line and the weight of any fluid in the capillary line. The weight of the weight bar **226** of the dynamic valve assembly **200** can be selectively adjusted to increase or decrease the weight of the dynamic valve assembly, thereby increasing or decreasing the force with which the dynamic valve assembly **200** is pushed into engagement with the stationary valve assembly.

(57) As the dynamic valve assembly **200** and capillary line are lowered down the production tube **117**, the bottom of the dynamic valve assembly **200** approaches the dart **120** at the top of the stationary valve assembly **100**, as illustrated in FIG. **7**A. As the dynamic valve assembly **200** and the capillary line are lowered further down the production tube **117**, the tip **127** of the dart **120** enters into the interior space formed by the prongs **203** of the collet **202**, as illustrated in FIG. **7**B. As depicted in FIG. **2**, the inner surface of each of the prongs **203** of the collet **202** include leading the sloped surfaces **204**, inwardly projecting pads **201** and trailing sloped surfaces **206**. To the extent the collet **202** is not positioned in the exact center of the production tube **117**, the leading sloped surfaces **204** of the prongs **203** of the collet **202** help to center the collet **202** around the tip **127** of the dart **120**.

(58) FIG. **7**C illustrates the dynamic valve assembly lowered even further onto the dart **120**. As shown in FIG. **7**C, the tip **127** of the dart **120** has now entered into the interior space of the housing **208** of the dynamic valve assembly and is located inside the forward seals **210** provided on the interior surface of the housing **208**.

(59) FIG. **7**D illustrates the dynamic valve assembly lowered even further onto the dart **120**. At this point, the leading sloped surfaces **204** of the prongs **203** of the collet **202** have engaged the leading shoulder **123** of the annular ridge **122** of the dart **120**, which helps to further centralize the dynamic valve assembly with respect to the stationary valve assembly.

(60) Further downward motion of the dynamic valve assembly will cause the sloped surfaces **204** of the prongs **203** of the collet **202** to spread the prongs outward. The material of the collet **202** as well as the length, width and thickness of the prongs **203** of the collet **202** can be selectively adjusted to adjust the amount of force that is required to cause the prongs **203** to flex outward. In addition, one can adjust the slope angle of the leading sloped surfaces **204** so that different amounts of force are required to cause the prongs **203** to flex outward during engagement and latching.

(61) FIG. **7**E illustrates the dynamic valve assembly lowered even further onto the dart **120**. As illustrated in FIG. **7**E, the prongs **203** of the collet have flexed outward such that pads **201** of the prongs **203** of the collet **202** are now riding along the external surface of the annular ridge **122** of

the dart **120**. FIG. **7**E also illustrates that the exterior cylindrical surface of the dart **120** located just below the radial passageways **126** of the dart **120** have engaged the first of two seals **210** on the interior of the housing **208**.

(62) FIG. **7**F illustrates the dynamic valve assembly lowered even further onto the dart **120**. FIG. **7**F illustrates a condition where the tip **127** of the dart is approaching the ball **122** inside the housing **208**. In addition, both of the seals **210** on the interior of the housing **208** have now engaged the exterior cylindrical surface of the dart **120** below the radial fluid passageways **126**. As a result, a fluid tight seal has been formed between the fluid delivery passageway **205** inside the housing **208** and the exterior of the dart **120**.

(63) FIG. **7**G illustrates the dynamic valve assembly lowered further onto the dart **120**. As is apparent in FIG. **7**G, the pads **201** of the prongs **203** of the collet **202** are almost past the end of the annular ridge **122** on the exterior of the dart **120**. In addition, the tip **127** of the dart **120** has begun to contact the ball **212** inside the housing **208**.

(64) Further downward movement of the dynamic valve assembly causes the pads **201** of the prongs **203** of the collet **202** to move past the trailing shoulder **124** of the annular ridge **122** on the dart **120**. At this point, the prongs **203** of the collet **202** contract inward and trailing sloped surfaces **206** at the upper end of the pads **201** of the prongs **203** bear against the trailing shoulder **124** of the annular ridge **122** of the dart **120** to keep the dynamic valve assembly latched onto the dart **120**. In addition, the tip **127** of the dart **120** has pushed the ball **212** out of engagement with the valve seat **214** of the housing **208**. This slightly compresses the spring **213** which tends to bias the ball **122** into engagement with the valve seat **214**.

(65) With the ball **212** moved away from the seat **214**, fluid from the capillary line is free to flow down through the interior flow passageway **215** of the cap **216** into the fluid receiving passageway **231** of the housing **208**. The fluid can then flow around the ball **212** past the valve seat **214** and past the tip **127** of the dart **122**. Note, the diameter of the tip of the dart **122** is smaller than the interior diameter of the associated part of the housing **208**, so that fluid can flow through the annular space located between the exterior of the tip **127** of the dart **120** and the interior cylindrical passageway of the housing **208**.

(66) The fluid flows along the fluid delivery channel **205** of the housing **208**, and then through the radial fluid passageways **126** of the dart **120** into the axial fluid passageway **125** of the dart **120**. The fluid can then flow down through the axial fluid passageway **125** of the dart to the injection valve **110** of the stationary valve assembly **100**. The fluid can then be delivered from the injection valve **110** into the flow passageway **104** of the associated seating nipple. The fluid can then pass along the the radial passageway **106** of the seating nipple to an area in the wellbore which is at the exterior of the production tube **117**.

(67) When one wishes to disconnect the dynamic valve assembly **200** from the stationary valve assembly **100**, it is possible to simply draw the capillary line and the attached dynamic valve assembly **200** back up to the top of the wellbore. The amount of force required to unlatch the dynamic valve assembly **200** from the dart **120** of the stationary valve assembly **100** will depend on the flexibility of the prongs **203** on the collet **202** of the dynamic valve assembly, as well as the slope angle of the trailing sloped surfaces **206** of the pads **201** of the prongs **203**. One can adjust all these factors to selectively vary the amount of force required to unlatch the dynamic valve assembly **200** from the stationary valve assembly **100**.

(68) When the dynamic valve assembly **200** unlatches from the stationary valve assembly **100**, the tip **127** of the dart **120** will back away from the ball **212** in the interior of the housing **208** of the dynamic valve assembly **200**. The spring **213** will then push the ball **212** back into engagement with the valve seat **214**, thereby sealing off the fluid flow from the capillary line. This helps to prevent any fluid from the interior of the production tube **117** from entering the capillary line.

(69) In the embodiments described above, the stationary valve assembly includes a seating nipple **109**, an injection valve **110**, and the dart **120**. All of these elements are permanently mounted to the

bottom of the production tube **117**. In alternate embodiments some of these elements could be releasably mounted to the bottom of the production tube **117**. Also, in alternate embodiments, some of these elements could be moved to the dynamic valve assembly.

(70) FIG. **8** illustrates a seating coupler **150** that has been installed in a portion of a production tube **117***a*. As illustrated in FIG. **8**, a seating coupler **150** is mounted to a first portion of the production tube **117***a*, and the lower end of the seating coupler **150** is attached to the upper end of a collar **115**. The lower end of the collar **115** is attached to a second portion of the production tube **117***b*. The seating coupler **150** is configured such that one or more components can be detachably mounted to the seating coupler **150**.

(71) The concept here is to assemble the production tube as depicted in FIG. **8**. Later, when it is desirable to begin delivering a fluid or chemical into the well via a capillary line, all the components needed to deliver the fluid or chemical can be installed at the appropriate location within the production tube via wireline. If it later becomes desirable to remove all the components being used to deliver the fluid or chemical into the well, they can be retrieved via wireline.

(72) FIG. **9** illustrates the same section of production tubing **117***a*/**117***b* as depicted in FIG. **8**. However, in FIG. **9** a dart assembly has been mounted to the seating coupler **150**. The dart assembly includes a cylindrical housing **152**, an adaptor **140** attached to the cylindrical housing **152** and a dart **120** attached to the adaptor **140**. The dart assembly would be run down upper section of the production line via wireline and then mounted to the seating coupler **150**. If it later becomes desirable or necessary to remove the dart assembly, the dart assembly can be removed from the seating coupler via wireline.

(73) FIG. **10** illustrates a dynamic valve assembly that could be latched onto the dart assembly depicted in FIG. **9** after the dart assembly has itself been releasably mounted on the seating coupler **150**. As shown in FIG. **10**, an injection valve **110** is mounted between two weight bars **226**, which are themselves mounted inside centralizers **230**. The latching and unlatching mechanisms remain unchanged. Also, the function of the injection valve **110** in controlling the delivery of fluid into the well from the capillary line and preventing fluid flow back up into the capillary line remains unchanged. The only difference is that the injection valve **110** has been moved to be a part of the dynamic valve assembly.

(74) FIG. **11** depicts the dynamic valve assembly as illustrated in FIG. **10** mounted to the dart assembly that is itself attached to the seating coupler **150**. As described above, the dynamic valve assembly could be lowered into engagement with the dart **120** to operatively couple a capillary line to the dart **120**, as depicted in FIG. **11**. The process of latching the dynamic valve assembly depicted in FIG. **10** to the dart assembly would proceed substantially the same as described above in connection with FIGS. **7**A-**7**H.

(75) In another alternate embodiment, the elements that are mounted to the seating coupler could include both the dart assembly and an injection valve **110**, as depicted in FIG. **12**. In this embodiment, a connector **147** that is attached to both the dart **120** and the injection valve **110** is mounted to the a cylindrical housing **152**. The cylindrical housing is then mounted to the seating coupler via wireline. Here again, it if becomes desirable or necessary, one would remove that entire assembly via wireline.

(76) The dart and injection valve assembly depicted in FIG. **12** could be used in connection with a dynamic valve assembly **200** like the one depicted above in FIG. **2**. The dynamic valve assembly **200** would be latched onto the dart **120** in the same way described above.

(77) The specific features of the various elements of the dynamic valve assembly and the stationary valve assembly in foregoing embodiments are only examples. One could make many modifications to those elements and achieve the same overall functionality. Thus, the descriptions provided above of specific embodiments should in no way be considered limiting.

(78) For example, the ball **212** and seat **214** in the housing **208** of the dynamic valve assembly **200** could be replaced with a different type of actuation mechanism. For example, a dart valve having a

stem and a seating surface could replace the ball **212**. The stem of the dart valve could extend down into the fluid delivery passageway **205** of the housing. In this case, a stationary element of the stationary valve assembly could push the dart valve upward as the dynamic valve assembly is lowered within the production tubing to open the valve assembly and allow pressurized fluid to flow from the dynamic valve assembly into the stationary valve assembly.
(79) As another example, in the foregoing embodiments the prongs **203** of a collet **202** latch to an outwardly protruding annular ridge **122** of the dart **120** to latch the dynamic valve assembly to the stationary valve assembly. In alternate embodiments a different sort of latching arrangement could be provided. Those of skill in this art are aware of multiple different ways of releasably latching a first tool element to a second tool element within a wellbore, and any such alternate latching arrangement could be used.
(80) FIG. **13** is a sectional view illustrating a portion of an alternate embodiment of a dynamic valve assembly that can be used to deliver pressurized fluid into a well. FIG. **13** illustrates a spear **300** that would be connected to the end of a capillary line which is run down into a well bore to deliver a fluid from the capillary line into a portion of the well bore. The spear **300** includes a first end **304** and a second end **302**. The second end **302** would be attached to the capillary line used to deliver fluid into the well bore.
(81) The spear **300** includes a cylindrical hollow body with a plurality of radially extending fluid supply passageways **310** located adjacent to the first end **304** of the spear **300**. The radial fluid supply passageways **310** would emit the fluid delivered through the capillary line to the interior of the hollow body of the spear **300**.
(82) A first cylindrical sealing member **306** is provided on the outer cylindrical surface of the spear **300** on a first side of the fluid supply passageways **310**. A second cylindrical sealing member **308** is provided on the outer cylindrical surface of the spear **300** on the other side of the fluid supply passageways **310**. The first and second cylindrical sealing members **306**, **308** are configured to form a seal with a flow plug as described in greater detail below.
(83) FIG. **14** illustrates the spear **300** connected to the end of a dynamic valve assembly that would be attached to a capillary fluid supply line. The assembly includes an injection valve **110** mounted between two weight bars **226**, which are themselves mounted inside centralizers **230**. The spear **300** as depicted in FIG. **13** is then attached to the lower end of the second weight bar **226**.
(84) FIGS. **15**A and **15**B illustrate parts of a flow plug assembly **330** which would be mounted at a particular location within the wellbore. Typically the flow plug assembly **330** would be mounted to a production tube within the wellbore at a location where fluid from a capillary supply line is to be delivered into the well.
(85) The flow plug assembly **330** includes a hollow cylindrical body **340** which is attached to an x-lock assembly **350**. Internal threads **346** on a coupling portion **344** at a second end of the hollow cylindrical body **340** are screwed onto external threads **352** on the x-lock assembly **350**. FIG. **15**A shows the two parts prior to assembly and FIG. **15**B shows the two parts once they have been assembled together at the coupling **344**. The hollow cylindrical body **340** of the flow plug assembly **330** includes a plurality of radially extending flow passageways **342**. The x-lock assembly **350** also includes a fishing neck **354** at a second end **356**. The flow plug assembly **330** depicted in FIGS. **15**A and **15**B is designed to receive a spear as depicted in FIG. **13**.
(86) Alternate embodiments could include similar types of downhole positioning tools. Thus, the use of an x-lock assembly in this embodiment should in no way be considered limiting of the way in which a flow plug assembly could be created. Alternate embodiments could include a different type of positioning tool. Or, no positioning tool elements whatsoever.
(87) FIG. **16** illustrates the combination of the spear **300** depicted in FIG. **13** and the flow plug assembly **330** depicted in FIGS. **15**A and **15**B. As shown in FIG. **16**, the first end **304** of the spear **300** is inserted into the interior of the hollow cylindrical body **340** of the flow plug assembly **330** so that the radially extending fluid supply passageways **310** of the spear **300** are aligned with the

flow passageways **342** on the hollow cylindrical body **340** of the flow plug assembly **330**. When the second end **302** of the spear **300** is attached to the end of a capillary fluid supply line, fluid is delivered in the interior of the spear **300**. The fluid can then flow through the fluid supply passageways **310** of the spear **300** and then out the flow passageways **342** of the flow plug assembly **330** to deliver the fluid from the capillary supply line to the interior of the wellbore.

(88) FIG. **16** also illustrates that the first and second sealing members **306**, **308** on the external cylindrical surface of the spear **300** bear against and form a fluid tight seal with the interior cylindrical wall of the hollow cylindrical body **340** of the flow plug assembly **330**. This ensures that the fluid delivered to the interior bore of the spear **300** is emitted into the wellbore, rather than just to the interior of the flow plug assembly **330**.

(89) FIG. **17** is a sectional view of another embodiment of a flow plug assembly **332** which can be part of a dynamic valve assembly for delivering fluid into a well. In this embodiment, a hollow cylindrical shutter **420** is mounted in the interior of a hollow cylindrical body **400** of the flow plug assembly **332** A biasing member **410** trapped between a first end of the hollow cylindrical shutter **420** and the first end of the interior of the hollow cylindrical body **400** of the flow plug assembly **332**. The biasing member **410** biases the hollow cylindrical shutter **420** into a closed position. In the closed position, the hollow cylindrical shutter **420** closes off the radially extending flow passageways **442** of the flow plug assembly **332**.

(90) A plurality of circular seals **422** and **424** are mounted around the exterior of the hollow cylindrical shutter **420**. The circular seals **422**, **424** form a fluid tight seal between the exterior of the hollow cylindrical shutter **420** and the interior cylindrical surface of the hollow cylindrical body **400** of the flow plug assembly **332**. In alternate embodiments, different types of sealing elements could be used.

(91) When the hollow cylindrical shutter **420** is in the closed position, as illustrated in FIG. **17**, a first set of circular seals **422** are located on a first side of the radially extending flow passageways **442** and the second set of circular seals **424** are located on the opposite side of the radially extending passageways **442**. This ensures that the interior of the flow plug assembly **332** is sealed off from the radially extending flow passageways **442** such that any fluid in the interior of the flow plug assembly **332** cannot escape through the flow passageways **442**, and wellbore fluid cannot ingress into the tubing section.

(92) FIG. **18**A illustrates the flow plug assembly **332** depicted in FIG. **17** with a spear **300** partially inserted into the flow plug assembly **332**. FIG. **18**B shows the spear **300** full inserted into the flow plug assembly **332**. As shown in FIG. **18**A, as the spear **300** is inserted into the interior of the flow plug assembly **332**, a first end **304** of the spear **300** abuts the second end of the hollow cylindrical shutter **420**. As the spear **300** is inserted further into the flow plug assembly **332**, the first end **304** of the spear **300** pushes the hollow cylindrical shutter **420** downward against the biasing action of the biasing member **410**. When the spear **300** is fully inserted into the flow plug assembly **332**, the radially extending fluid supply passageways **310** of the spear **300** align with the radially extending flow passageways **442** of the flow plug assembly **332** so that fluid from the interior of the spear **300** can be emitted out of the flow passageways **442** of the flow plug assembly **332** and into the wellbore. As depicted in FIG. **18**B, when the spear **300** is fully inserted into the flow plug assembly **332**, the first and second seal members **306**, **308** on the exterior of the spear **300** are positioned on opposite sides of the radially extending flow passageways **442** of the flow plug assembly **332**.

(93) When the spear **300** is subsequently withdrawn from the flow plug assembly **332**, the biasing member **410** will push the hollow cylindrical shutter **420** back into the closed position depicted in FIGS. **17** and **18**A so that the hollow cylindrical shutter **420** will once again cover and close of the radially extending flow passageways **442**.

(94) FIGS. **19**A and **19**B depict an alternate embodiment of the flow plug assembly **334** which includes a post **460** located in the interior of the hollow cylindrical body **400** of the flow plug assembly **334**. In this embodiment, the biasing member **410** in the form of a coil spring occupies an

annular space formed between the exterior of the post **460** and the interior cylindrical surface of the hollow cylindrical body **400** of the flow plug assembly **334**. FIG. **19**A illustrates the spear **300** after it has been partially inserted into the flow plug assembly **334**. FIG. **19**B illustrates the spear **300** fully inserted into the flow plug assembly **334** such that the hollow cylindrical shutter **420** has been moved into the open position. As depicted in FIGS. **19**A and **19**B, the hollow cylindrical shutter **420** also slides within the annular space between the exterior of the post **460** and the interior cylindrical surface of the hollow cylindrical body **400** of the flow plug assembly **334**. Thus, the post helps to support and guide the movement of the hollow cylindrical shutter **420**.
(95) FIG. **20** illustrates another alternate embodiment of the spear **500** which includes a hollow cylindrical shutter **520** mounted around the exterior cylindrical surface of the spear **500**. In this embodiment, a biasing member in the form of a coil spring **530** is positioned between a shoulder **507** on the exterior of the spear **500** and a second end **524** of the hollow cylindrical shutter **520**. The biasing member **530** biases the hollow cylindrical shutter **520** into a closed position, as depicted in FIG. **20**. When the hollow cylindrical shutter **520** is in the closed position, it covers the radially extending fluid supply passageways **510**.
(96) The spear **500** in this embodiment still includes first and second sealing members **506**, **508** located on opposite sides of the radially extending fluid supply passageways **510**. The first and second sealing members **506**, **508** provide a fluid tight seal between the exterior of the spear **500** and the interior of the hollow cylindrical shutter **520**. When the hollow cylindrical shutter **520** is in the closed position, as depicted in FIG. **20**, the first and second sealing members **506**, **508** prevent any fluid in the interior of the spear **500** from escaping through the radially extending fluid supply passageways **510**.
(97) FIG. **21** depicts a flow plug assembly **551** which could be used in conjunction with the spear depicted in FIG. **20**. This embodiment of the flow plug assembly **551** includes a plurality of radially extending flow passageways **552**. This embodiment of the flow plug assembly **551** also includes a shoulder **553** on the interior cylindrical surface of the flow plug assembly **551** that can be used to cause the hollow cylindrical shutter **520** on the exterior of the spear **500** to move from the closed position to an open position.
(98) FIGS. **22**A and **22**B show the combination of the spear **500** of FIG. **20** and the flow plug assembly **551** of FIG. **21** as the spear **500** is inserted into the flow plug assembly **551**. FIG. **22**A shows the spear **500** partially inserted into the flow plug assembly **551** so that a first end of the hollow cylindrical shutter **520** has been brought into engagement with the shoulder **553** of the flow plug assembly **551**. As the spear **500** is inserted further into the flow plug assembly **551**, the shoulder **553** on the flow plug assembly **551** causes the shutter to move to an open position as depicted in FIG. **22**B. When the hollow cylindrical shutter **520** is in the open position, vent passageways **525** on the hollow cylindrical shutter **520** align with the fluid supply passageways **510** of the spear **500** and with the flow passageways **552** of the flow plug assembly **551**. This allows fluid from the interior of the spear **500** to flow out through the fluid supply passageways **510**, through the vent passageways **525** of the hollow cylindrical shutter **520** and then through the flow passageways **552** of the flow plug **551** so that the fluid can be emitted into the wellbore.
(99) When the hollow cylindrical shutter **520** is in the open position, the biasing member **530** has been compressed. When the spear **500** is withdrawn from the interior of the flow plug assembly **551**, the biasing member **530** will cause the hollow cylindrical shutter **520** to return to the closed position, as depicted in FIG. **20**. As a result, fluid from the interior of the spear **500** will no longer be allowed to flow out of the fluid supply passageways **510** of the spear **500**.
(100) FIG. **23** illustrates an alternate embodiment of a flow plug assembly **551** similar to the one depicted in FIG. **21**. In the embodiment depicted in FIG. **23**, a check valve **570** is attached to the first end **554** of the flow plug assembly **551**. Interior threads **572** on the check valve engage exterior threads **559** on the second end **554** of the flow plug assembly **551** to attach the check valve **570** to the flow plug assembly **551**. A flow passageway **557** on the second end **554** of the flow plug

assembly **551** allows fluid in the interior of the flow plug assembly **551** to communicate with an interior of the check valve **570**.

(101) With the embodiment as depicted in FIG. **23**, as the spear **500** is inserted into the interior of the flow plug assembly **551**, any fluid in the interior of the of the flow plug **551** located ahead of the end of the spear **500** can be pushed out of the first end **554** of the flow plug assembly **551** through the check valve **570** to avoid hydraulic locking and insertion of the spear **500** into the interior of the flow plug assembly **551**.

(102) FIGS. **24** and **25** depict another embodiment of a spear **600** and a flow plug assembly **653** which are similar in function to the spear and flow plug assembly depicted in FIGS. **20** and **21**. In this embodiment, the hollow cylindrical shutter **620** includes a portion which extends beyond the first end **604** of the spear **600**. The extending portion of the cylindrical shutter **620** includes a radial latching member **626** which protrudes radially outward from the exterior surface of the hollow cylindrical shutter **620**. FIG. **24** shows the hollow cylindrical shutter **620** in the closed position, where the shutter **620** covers the fluid supply passageways **610** of the spear **600**.

(103) The flow plug assembly **653** depicted in FIG. **25** includes an annular aperture **654** formed on the inner cylindrical surface of the flow plug assembly **653**. When the spear **600** is inserted into the flow plug assembly **653**, the radial latching member **626** on the end of the hollow cylindrical shutter **620** is received in the annular aperture **654** of the flow plug assembly **653**.

(104) FIGS. **26**A and **26**B illustrate the spear **600** being inserted into the interior of the flow plug assembly **653**. In FIG. **26**A, the spear **600** is partially inserted into the interior of the flow plug assembly **653** such that the radial latching element **626** that projects outward from the end of the hollow cylindrical shutter **620** is seated in the annular aperture **654** of the flow plug assembly **653**. FIG. **26**B illustrates that once the spear **600** is fully inserted into the flow plug assembly **653**, the hollow cylindrical shutter **620** is moved into an open position where vent apertures **625** of the hollow cylindrical shutter **620** are aligned with the radially extending fluid supply passageways **610** of the spear **600**. The vent apertures **625** are also aligned with flow passageways **652** of the flow plug assembly **652**. As a result, fluid from the interior of the spear **600** can flow out of the fluid supply passageway **610** through the vent passageways **625** and then out through the radially extending flow passageways **652** of the flow plug assembly **653**.

(105) In this embodiment, as the spear is withdrawn from the interior of the flow plug assembly **653**, because the annular latching element **626** on the end of the cylindrical shutter **620** is lodged in the annular aperture **654** of the flow plug assembly **653**, the hollow cylindrical shutter **620** is pulled into the closed position in conjunction with the biasing force provided by the spring **630**. This ensures that the hollow cylindrical shutter **620** covers the fluid supply passageways **610** of the spear **600** once the spear **600** has been withdrawn from the flow plug assembly **653**.

(106) The leading angle and the wall thickness of the annular latching element **626** and the corresponding profile of the annular aperture **654** of the flow plug assembly **653** can be selectively varied to adjust the amount of force required to fully insert the spear **600** into the flow plug assembly **653**. Similarly, the lagging angle and wall thickness of the annular latching element **626** and the corresponding profile of the annular aperture **654** of the flow plug assembly **653** can be selectively varied to adjust the amount of force required to remove or unseat the spear **600** from the flow plug assembly **653**. Of course, the biasing element **630** also provides forces that must also be taken into account in adjusting for the desired insertion and removal forces.

(107) FIG. **27** illustrates another embodiment of a flow plug assembly **653** similar to the one depicted in FIG. **25**. In this embodiment, a dual direction check valve **680** is attached to the first end **662** of the flow plug assembly **653**. The dual direction check valve **680** includes a first fluid circuit which includes a first check valve **682** located between first and second fluid lines **683**, **681**. The dual direction check valve **680** also includes a second check valve **684** located between first and second fluid lines **685**, **686**. The dual direction check valve allow fluid to flow in both directions. However, it requires less force for the fluid to flow in a first direction than in a second

direction.
(108) The dual direction check valve **680** will operate in such a fashion that as the spear **600** is inserted into the flow plug assembly **653**, fluid in the interior of the flow plug assembly **653** can be easily pushed through the first circuit of the dual direction check valve **680** that includes the fluid passageways **681** and **683** and the valve **682**. In addition, as the spear **600** is withdrawn from the flow plug assembly **653**, fluid from the exterior of the check valve can be drawn into the flow plug assembly **653** through the second circuit of the dual direction check valve **680** that includes the flow passageways **685** and **686** and the valve **684** to make it easier to withdraw the spear **600** from the interior of the flow plug assembly than if the check valve **680** were not present. However, it will require more force to cause the second valve **684** to open than is required to cause the first valve **682** open. This can tend to keep the spear seated within the flow plug assembly **653**.
(109) The foregoing descriptions related to the delivery and retrieval of a downhole capillary line for injection of fluid into a well. However, the systems and methods described above could be used for the delivery and retrieval of other downhole tools, such as pressure gauges and sensors. Thus, the foregoing descriptions should in no way be considered limiting.
(110) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

Claims

1. An assembly for delivering a fluid into a well, comprising: a flow plug configured to be coupled to a production tube within a wellbore, the flow plug comprising: a hollow cylindrical body, having a closed end and an open end; and a plurality of flow passageways that extend through the cylindrical body in a radial direction, the flow passageways being located adjacent the closed end; and a spear that is configured to be inserted into an interior of the hollow cylindrical body of the flow plug, the spear comprising: a hollow cylindrical body having a first end and a second end that is configured to be attached to a fluid supply line such that fluid from the fluid supply line can flow into an interior of the hollow cylindrical body; a plurality of fluid supply passageways that extend through the hollow cylindrical body in a radial direction, the plurality of fluid supply passageways being located such that when the spear is fully inserted into the hollow cylindrical body of the flow plug, fluid in the interior of the hollow cylindrical body of the spear can flow out of the fluid supply passageways and into the flow passageways of the flow plug; a first sealing member located on an exterior of the hollow cylindrical body at a location between the plurality of fluid supply passageways and the second end of the hollow cylindrical body, the first sealing member being configured to form a seal between the exterior of the spear and an interior of the flow plug.
2. The assembly of claim 1, wherein the spear further comprises a second sealing member located on an exterior of the hollow cylindrical body at a location between the plurality of fluid supply passageways and the first end of the hollow cylindrical body of the spear, the second sealing member being configured to form a seal between the exterior of the spear and an interior of the flow plug.
3. The assembly of claim 1, wherein the flow plug further comprises a hollow cylindrical shutter that is slidably mounted in the interior of the hollow cylindrical body of the flow plug such that the shutter can move between a closed position at which the shutter blocks off the flow passageways and an open position at which the shutter exposes the flow passageways.
4. The assembly of claim 3, further comprising at least one circular seal positioned between an

exterior cylindrical surface of the shutter and an interior cylindrical surface of hollow cylindrical body of the flow plug, wherein the at least one circular seal forms a fluid-tight seal between the exterior cylindrical surface of the shutter and the interior cylindrical surface of hollow cylindrical body of the flow plug.
5. The assembly of claim 3, further comprising a biasing member located in the interior of the hollow cylindrical body of the flow plug, wherein the biasing member biases the shutter to the closed position, and wherein when the spear is inserted into the flow plug, a bottom end of the spear pushes the shutter from the closed position to the open position against a biasing force of the biasing member.
6. The assembly of claim 5, wherein the biasing member comprises a coil spring, wherein the flow plug includes a post located at the closed end of the hollow cylindrical body, the post extending along a central longitudinal axis of the flow plus, and wherein the coil spring is positioned in an annular space between an exterior of the post and an interior cylindrical surface of the flow plug.
7. The assembly of claim 1, wherein the spear further comprises a hollow cylindrical shutter that is slidably mounted on the exterior of the hollow cylindrical body of the spear such that the shutter can move between a closed position at which the shutter covers the fluid supply passageways and an open position at which vent passageways of the shutter align with and expose the fluid supply passageways.
8. The assembly of claim 7, further comprising at least one circular seal located on an exterior cylindrical surface of the shutter, the at least one circular seal being configured to provide a fluid-tight seal between the exterior cylindrical surface of the shutter and an interior cylindrical surface of hollow cylindrical body of the flow plug.
9. The assembly of claim 8, wherein the first sealing member forms a seal between an exterior of the spear and an interior cylindrical surface of the shutter.
10. The assembly of claim 7, further comprising a biasing member located on an exterior of the hollow cylindrical body of the spear, wherein the biasing member biases the shutter to the closed position, and wherein when the spear is inserted into the flow plug, a shoulder on an inner cylindrical surface of the flow plug bears against a first end of the shutter and pushes the shutter from the closed position to the open position against a biasing force of the biasing member.
11. The assembly of claim 10, wherein the biasing member comprises a coil spring that surrounds the exterior of the spear, and wherein the coil spring is located between a second end of the shutter and a shoulder formed on the exterior of the spear.
12. The assembly of claim 7, wherein a first end of the hollow cylindrical shutter includes a latching element that projects radially outward from the first end of the hollow cylindrical shutter, wherein a cylindrical aperture is formed on an inner cylindrical surface of the hollow cylindrical body of the flow plug, and wherein when the spear is inserted into the flow plug, the latching element of the hollow cylindrical shutter is received in the cylindrical aperture of the flow plug.
13. The assembly of claim 12, wherein when the hollow cylindrical shutter is located in the closed position, a portion of the hollow cylindrical shutter having the latching element projects beyond the first end of the hollow cylindrical body of the spear.
14. The assembly of claim 1, further comprising a valve that is attached to the closed end of the hollow cylindrical body of the flow plug, wherein the valve is in fluid communication with an interior of the hollow cylindrical body of the flow plug.
15. The assembly of claim 14, wherein the valve is a check valve.
16. The assembly of claim 14, wherein the valve is a two-way check valve.
17. A spear that is configured to be attached to a fluid supply line and to deliver fluid from the fluid supply line into a flow plug mounted on a production tube within a wellbore, comprising: a hollow cylindrical body having a first end and a second end that is configured to be attached to a fluid supply line such that fluid from the fluid supply line can flow into an interior of the hollow cylindrical body; a plurality of fluid supply passageways that extend through the hollow cylindrical

body in a radial direction, the plurality of fluid supply passageways being located adjacent the first end of the hollow cylindrical body; a first sealing member located on an exterior of the hollow cylindrical body at a location between the plurality of fluid supply passageways and the second end of the hollow cylindrical body; and a hollow cylindrical shutter having vent passageways that is slidably mounted on the exterior of the hollow cylindrical body such that the shutter can move between a closed position at which the shutter covers the fluid supply passageways and an open position at which the vent passageways align with and expose the fluid supply passageways, and wherein the first sealing member forms a seal between the exterior of the hollow cylindrical body and an interior cylindrical surface of the shutter.

18. The assembly of claim 17, wherein the spear further comprises a second sealing member located on the exterior of the hollow cylindrical body at a location between the plurality of fluid supply passageways and the first end of the hollow cylindrical body, the second sealing member being configured to form a seal between the exterior of the hollow cylindrical body and the interior cylindrical surface of the shutter.

19. The assembly of claim 17, further comprising at least one circular seal located on an exterior cylindrical surface of the shutter, the at least one circular seal being configured to provide a fluid-tight seal between the exterior cylindrical surface of the shutter and an interior cylindrical surface of a flow plug.

20. The assembly of claim 17, further comprising a biasing member located on an exterior of the hollow cylindrical body, wherein the biasing member biases the shutter to the closed position, and wherein the shutter is configured such that when the spear is inserted into a flow plug, a shoulder on an inner cylindrical surface of the flow plug can bear against a first end of the shutter and push the shutter from the closed position to the open position against a biasing force of the biasing member.

21. The assembly of claim 20, wherein the biasing member comprises a coil spring that surrounds the exterior of the hollow cylindrical body, and wherein the coil spring is located between a second end of the shutter and a shoulder that protrudes from the exterior of the hollow cylindrical body.

22. The assembly of claim 17, wherein a first end of the hollow cylindrical shutter includes a latching element that projects radially outward from the first end of the hollow cylindrical shutter, wherein the latching element is configured such that when the spear is inserted into a flow plug, the latching element of the hollow cylindrical shutter is received in a cylindrical aperture of the flow plug.

23. The assembly of claim 22, wherein when the hollow cylindrical shutter is located in the closed position, a portion of the hollow cylindrical shutter having the latching element projects beyond the first end of the hollow cylindrical body of the spear.